Yuxuan Ni

(412) 618-9972 **|** niyx25@gmail.com**|** https://www.linkedin.com/in/yuxuanni/

EDUCATION

Carnegie Mellon University **Jan 2023- Dec 2023**

MSc in Engineering & Technology Innovation Management *Pittsburgh, PA*

University of Electronic Science and Technology of China (UESTC) **Sep 2018- Jun 2022**

*B.E. in Communication Engineering from **UESTC*** *Chengdu, China*

*B.E in Electronics and Electrical Engineering with Communications from the **University of Glasgow***

PUBLICATION

Y. Ni and L. Shi, "Specific clustering scenarios and simple algorithm simulation in 5G environment," Proc. SPIE 12087, International Conference on Electronic Information Engineering and Computer Technology (EIECT 2021), 120870M (13 December 2021); https://doi.org/10.1117/12.2624711

RESEARCH EXPERIENCE

MVMD-Based CSP for EEG Signal Classification **Mar 2024- Present**

Research assistant, supervised by Prof. Aimin Jiang from Hohai University

- Applied **Multivariate Variational Mode Decomposition (MVMD)** on **over 1,000 datasets** to validate inter- and intra-subject EEG signal variability.
- Designed an algorithm based on **Common Spatial Pattern (CSP)**, achieving an around **10%** improvement in classification accuracy.
- Integrated **machine learning** techniques to reduce computational complexity by about **20%**.
- Optimized spatial and mode filters using the **Alternate Direction Method of Multipliers (ADMM).**
- Leveraged **Multilayer Perceptron (MLP)** for signal classification, outperforming traditional Linear Discriminant Analysis (LDA) and Support Vector Machines (SVM) in accuracy.

NGSO Satellite In-line Interference Analysis **Aug 2023- Jan 2024**

Research assistant, supervised by Prof. Jon Peha and Prof. Marvin Sirbu from CMU

- Simulated communication links between satellite constellations and ground stations using **MATLAB** and **Python** with TLE data, achieving location high accuracy within **1 km**.
- Developed and refined a custom simulation model capable of efficiently processing **million-level** datasets to assess in-line interference among various satellite constellations.
- Enhanced simulation coverage by incorporating diverse **satellite communication models**, reducing computational cost by around **60%**.
- Designed and tested an interference avoidance algorithm, decreasing interference probabilities by **30%**.

Low-Noise Balanced Photodetector for Single Photon Detection **Sep 2021- May 2022**

Research assistant, supervised by Prof. Yunxiang Wang from UESTC

- Reduced noise and enhanced sensitivity by integrating JFETs with photodiode circuits, achieving a **300 MHz** bandwidth.
- Conducted simulations using **LTspice**, evaluating **8 circuit designs** for optimal performance.
- Designed and optimized **PCB** layouts in **Altium Designer**, effectively minimizing **oscillation** and improving circuit performance.

5G Clustering Algorithm Simulation for Ultra-Dense Network **May 2021- Oct 2021**

Research assistant, supervised by Prof. Yuwei Wang from Chinese Academy of Sciences

- Developed an innovative clustering algorithm tailored for 5G ultra-dense networks, integrating **path loss** mitigation and **frequency management** strategies to minimize serial interference.
- Validated algorithm performance through extensive **NS3** simulations, achieving a **30%** improvement in data reception rates.

Design and Construction of a Multi-Function Rover **Feb 2021- Jun 2021**

Team project, leader of communication group

- Led the development of a low-latency wireless communication system for a multifunctional rover, ensuring seamless data exchange **with <50 ms latency**.
- Implemented advanced route tracking and color recognition capabilities using cutting-edge sensor technology and custom algorithms, achieving **over 90% navigation accuracy**.

WORK EXPERIENCE

Center for Technology Transfer and Enterprise Creation, CMU **May 2023- Aug 2023**

Enterprise creation, intern *Pittsburgh, PA*

Leader of market research team, CMU Bio-robotic Lab

- Built detailed profiles for **over 1,000** customers, identified **60+** high-potential leads, and established connections with **26** key prospects, driving a **20%** projected improvement in customer satisfaction.
- Analyzed over **one million** data points using **RStudio** and **Excel** to evaluate market potential, aligning technical solutions and business models with evolving customer demands.

Leader of consultant team, Phlux

- Spearheaded market trend analyses and conducted **20+** customer interviews to refine product use cases.
- Defined **over 10** proprietary technology use cases, helping Phlux reposition its product portfolio by leveraging insights from comparative market feedback.

Signify (China) Investment Co., Ltd. **Sep 2022- Dec 2022**

Hardware engineer, intern *Shanghai, China*

- Optimized power circuit designs by employing active **EMI filters** and high-performance **capacitors**, reducing power consumption by **30%**.
- Debugged and tested **IoT lamp** functionalities, enhancing lighting **accuracy** and user **response speed** by **50%** through refined control algorithms and logic.

China Mobile Communication Group Co., Ltd. **Jul 2020- Aug 2020**

Engineering maintenance engineer, intern *Changzhou, China*

- Strategically developed and deployed **10** 5G base stations, expanding network coverage by **20%** in critical areas.
- Improved operational efficiency and minimized network downtime, achieving a **15%** reduction in annual maintenance costs.

SKILLS

- Application Software: MATLAB, Keli uVision5, NS3, LTspice, Altium Designer
- Programming Languages: C, VHDL, Python, SQL, R